Readme · MD

📚 Student Study Planner

A console-based Java application to help students organize their subjects, manage study tasks, set priorities and deadlines, and track their overall progress — all from the terminal.

🎯 Features

- ➕ Add and delete **subjects**
- ✅ Add, delete, and manage **tasks** per subject
- 🔴 Set task **priority** — HIGH, MEDIUM, or LOW
- 📅 Set task **deadlines** (DD-MM-YYYY format)
- ✔️ Mark tasks as **completed**
- 🔍 **Filter** tasks by priority
- 📊 View **progress** with a visual progress bar

🗂️ Project Structure

```
StudyPlanner/
│
├── src/
│   ├── Main.java       # Entry point — console menu and input handling
│   ├── Planner.java    # Core logic — manages subjects and tasks
│   ├── Subject.java    # Subject model — holds a list of tasks
│   └── Task.java       # Task model — stores task details
│
└── README.md
```

⚙️ Prerequisites

- Java JDK 8 or higher
- A terminal / command prompt

🚀 How to Run

1. Clone the repository

```bash
git clone https://github.com/YOUR_USERNAME/StudyPlanner.git
cd StudyPlanner
```

2. Compile the source files

```bash
javac src/*.java
```

3. Run the application

```bash
java src.Main
```

🖥️ Usage

When you run the app, you'll see this menu:

```
╔═══════════════════════════════════════╗
  STUDENT STUDY PLANNER v1.0      ║
   Organize. Focus. Succeed.      ║
╚═══════════════════════════════════════╝

──────────────── MAIN MENU ────────────────
 SUBJECTS
 1. Add Subject
 2. Delete Subject
 3. View All Subjects
 TASKS
 4. Add Task
 5. Delete Task
 6. View All Tasks
 7. View Tasks by Subject
 8. Mark Task as Done
 9. Filter Tasks by Priority
 PROGRESS
 10. Show Progress
 0. Exit
```

Example Walkthrough

```
Enter choice: 1
Subject name: Mathematics

Enter choice: 4
```

```
Subject name: Mathematics
Task description: Complete Chapter 5 exercises
Deadline (DD-MM-YYYY): 05-04-2025
Priority (HIGH / MEDIUM / LOW): HIGH

Enter choice: 10

=== Study Progress ===
Subject: Mathematics        | Total: 1 | Done: 0 | Pending: 1
──────────────────────────────────────────────────────────────────────────

OVERALL: 0/1 tasks done  (0% complete)
[                                        ] 0%
```

🧠 Java Concepts Used

Concept	Where Applied
Classes & Objects	Task, Subject, Planner
Encapsulation	Private fields with public getters
Inheritance-ready design	Modular class hierarchy
ArrayLists	Storing subjects and tasks
Loops & Conditionals	Menu handling, task search
Static ID counter	Auto-incrementing task IDs

👤 Author

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